Started on	Monday, 16 May 2022, 2:29 PM
State	Finished
Completed on	Thursday, 19 May 2022, 3:13 PM
Time taken	3 days
Grade	**1.05** out of 1.50 (**70**%)

Question **1**

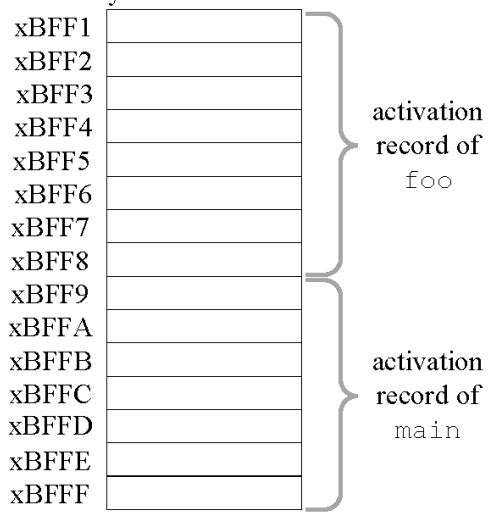
Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
void foo(int x, int* y);
int main() {
  int a, b;
  int *c, *d;
  a = 0x3;
  b = 0x4;
  c = &a;
  d = &b;
  foo(a,&b);
  printf("a=%d,  b=%d\n",a,b);
}
void foo(int x, int* y){
  int s, t;
  int *u;
  s = 0x1;
  t = 0x2;
  u = y;
  x = s - 1;
  *u = t;
}
```

The diagram below shows the memory of the machine executing the above program. The address of each memory location is shown next to each location.



Which one of the following statements is correct?

- a. The execution of statement "b = 0x4" in function main sets the value at location 0xBFFA to 0x4.
- b. The execution of statement "b = 0x4" in function main sets the value at location 0xBFFF to 0x4.
- c. The execution of statement "a = 0x3" in function main sets the value at location 0xBFFC to 0x3. ✔
- d. The execution of statement "a = 0x3" in function main sets the value at location 0xBFF9 to 0x3.

Your answer is correct.

The correct answer is:
The execution of statement “a = 0x3” in function main sets the value at location 0xBFFC to 0x3.

Question **2**

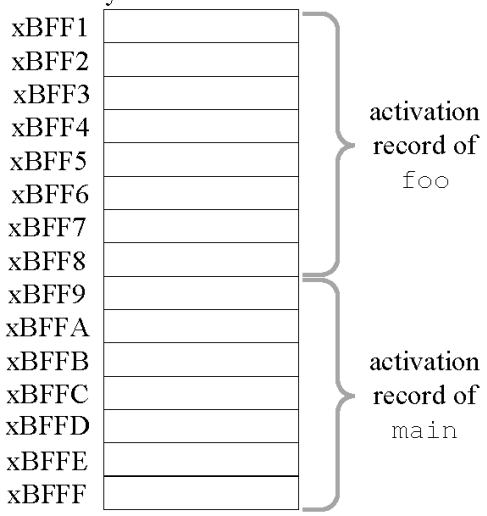
Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
void foo(int x, int* y);
int main() {
  int a, b;
  int *c, *d;
  a = 0x3;
  b = 0x4;
  c = &a;
  d = &b;
  foo(a,&b);
  printf("a=%d,  b=%d\n",a,b);
}
void foo(int x, int* y){
  int s, t;
  int *u;
  s = 0x1;
  t = 0x2;
  u = y;
  x = s - 1;
  *u = t;
}
```

The diagram below shows the memory of the machine executing the above program. The address of each memory location is shown next to each location.



Which one of the following statements is correct?

- a. The execution of statement "c = &a" in function main sets the value at location 0xBFFB to 0xBFF9.
- b. The execution of statement "c = &a" in function main sets the value at location 0xBFFB to 0x3.
- c. The execution of statement "c = &a" in function main sets the value at location 0xBFFE to 0xBFFF.
- d. The execution of statement "c = &a" in function main sets the value at location 0xBFFA to 0xBFFC. ✔

Your answer is correct.

The correct answer is:
The execution of statement “c = &a” in function main sets the value at location 0xBFFA to 0xBFFC.

Question **3**

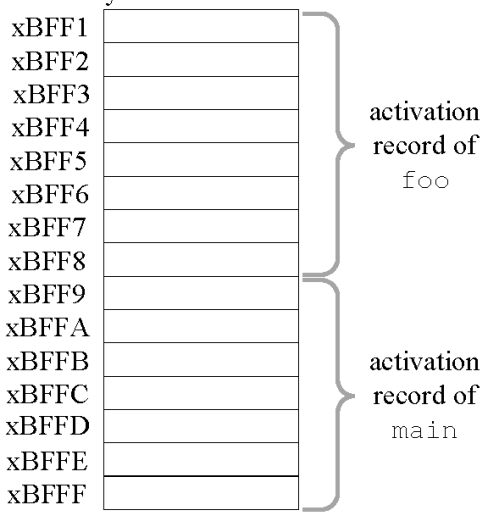
Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
void foo(int x, int* y);
int main() {
  int a, b;
  int *c, *d;
  a = 0x3;
  b = 0x4;
  c = &a;
  d = &b;
  foo(a,&b);
  printf("a=%d,  b=%d\n",a,b);
}
void foo(int x, int* y){
  int s, t;
  int *u;
  s = 0x1;
  t = 0x2;
  u = y;
  x = s - 1;
  *u = t;
}
```

The diagram below shows the memory of the machine executing the above program. The address of each memory location is shown next to each location.



Which one of the following statements is correct?

- a. During the execution of statement "*u = t" in function foo, the value in register 4 is 0xBFF3.
- b. During the execution of statement "d = &b" in function main, the value in register 5 is 0xBFF3.
- c. During the execution of statement "*u = t" in function foo, the value in register 7 is 0xBFF5.
- d. During the execution of statement "d = &b" in function main, the value in register 6 is 0xBFF9. ✔

Your answer is correct.

The correct answer is:
During the execution of statement “d = &b” in function main, the value in register 6 is 0xBFF9.

Question **4**

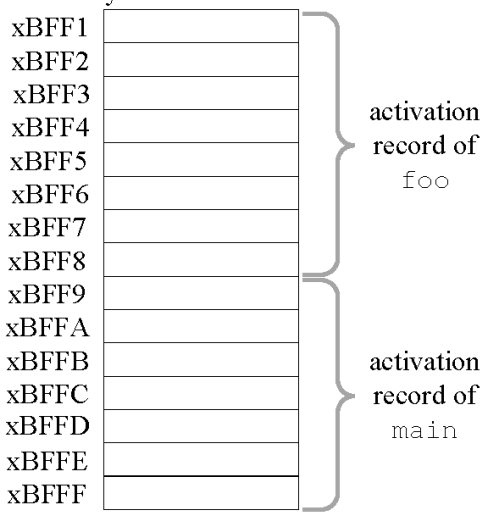
Incorrect

Mark 0.00 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
void foo(int x, int* y);
int main() {
  int a, b;
  int *c, *d;
  a = 0x3;
  b = 0x4;
  c = &a;
  d = &b;
  foo(a,&b);
  printf("a=%d,  b=%d\n",a,b);
}
void foo(int x, int* y){
  int s, t;
  int *u;
  s = 0x1;
  t = 0x2;
  u = y;
  x = s - 1;
  *u = t;
}
```

The diagram below shows the memory of the machine executing the above program. The address of each memory location is shown next to each location.



Which one of the following statements is correct?

- a. The execution of statement "*u = t" in function foo sets the value at location 0xBFF1 to 0x2. ✖
- b. During the execution of statement "*u = t" in function foo, the value at location 0xBFF4 is 0xBFFD.
- c. The execution of statement "*u = t" in function foo sets the value at location 0xBFF8 to 0x2.
- d. During the execution of statement "*u = t" in function foo, the value at location 0xBFF4 is 0xBFFC.

Your answer is incorrect.

The correct answer is:
During the execution of statement “*u = t” in function foo, the value at location 0xBFF4 is 0xBFFC.

Question **5**

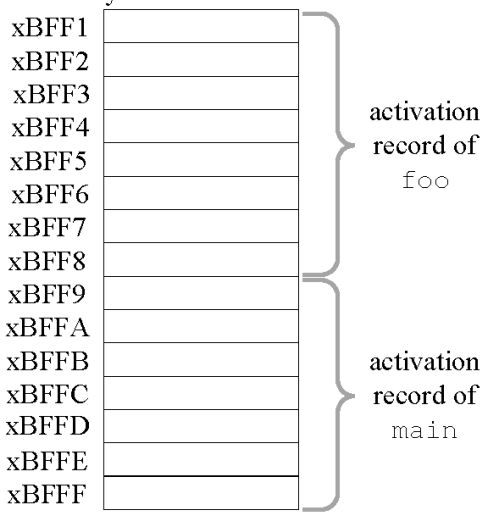
Incorrect

Mark 0.00 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
void foo(int x, int* y);
int main() {
  int a, b;
  int *c, *d;
  a = 0x3;
  b = 0x4;
  c = &a;
  d = &b;
  foo(a,&b);
  printf("a=%d,  b=%d\n",a,b);
}
void foo(int x, int* y){
  int s, t;
  int *u;
  s = 0x1;
  t = 0x2;
  u = y;
  x = s - 1;
  *u = t;
}
```

The diagram below shows the memory of the machine executing the above program. The address of each memory location is shown next to each location.



Which one of the following statements is correct?

- a. The execution of statement "u = y" in function foo sets the value at location 0xBFF1 to 0xBFF8. ✖
- b. The execution of statement "x = s - 1" in function foo sets the value at location 0xBFF8 to 0x0.
- c. The execution of statement "u = y" in function foo sets the value at location 0xBFF1 to 0xBFFB.
- d. The execution of statement "x = s - 1" in function foo sets the value at location 0xBFFC to 0x0.

Your answer is incorrect.

The correct answer is:
The execution of statement "u = y" in function foo sets the value at location 0xBFF1 to 0xBFFB.

Question **6**

Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
int main() {
  char* x;
  char y;
  char z[4] = "abc";
  x = z;
  y = z[2];
  z[0] = *(x+1);
  printf("%s\n",z);
}
```

Assume that (i) `rX` represents the register pointing to the beginning of the locations where the variables are stored, and (ii) the offsets of the variables in the program above are given in the table below. In the table, each capital letter represents an integer.

variable	offset
x	X
y	Y
z	Z

Which one of the following statements is correct?

- a. "x = z" is the same as "x = z[0]".
- b. "x = z" is the same as "x = *z".
- c. "x = z" is the same as "x = &z".
- d. "x = z" is the same as "x = &z[0]". ✔

Your answer is correct.

The correct answer is:
"x = z" is the same as "x = &z[0]".

Question **7**

Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
int main() {
  char* x;
  char y;
  char z[4] = "abc";
  x = z;
  y = z[2];
  z[0] = *(x+1);
  printf("%s\n",z);
}
```

Assume that (i) `rX` represents the register pointing to the beginning of the locations where the variables are stored, and (ii) the offsets of the variables in the program above are given in the table below. In the table, each capital letter represents an integer.

variable	offset
x	X
y	Y
z	Z

What is the offset value Z?

- a. -3
- b. -5 ✔
- c. 0
- d. -4

Your answer is correct.

The correct answer is:
-5

Question **8**

Incorrect

Mark 0.00 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
int main() {
  char* x;
  char y;
  char z[4] = "abc";
  x = z;
  y = z[2];
  z[0] = *(x+1);
  printf("%s\n",z);
}
```

Assume that (i) `rX` represents the register pointing to the beginning of the locations where the variables are stored, and (ii) the offsets of the variables in the program above are given in the table below. In the table, each capital letter represents an integer.

variable	offset
x	X
y	Y
z	Z

Which one of the following LC-3 instruction sequences will be generated when converting statement "x = z" to LC-3 assembly language instructions?

a.
```
ADD  R0, R5, #Z
STR  R0, R5, #X
```

b.
```
LDR  R0, R5, #Z
STR  R0, R5, #X
```

c.
```
ADD  R0, R5, #Z
LDR  R1, R0, #0
STR  R1, R5, #X
```

d.
```
LDR  R0, R5, #Z
LDR  R1, R0, #0
STR  R1, R5, #X
```

Your answer is incorrect.

The correct answer is:
ADD R0, R5, #Z
STR R0, R5, #X

Question **9**

Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
int main() {
  char* x;
  char y;
  char z[4] = "abc";
  x = z;
  y = z[2];
  z[0] = *(x+1);
  printf("%s\n",z);
}
```

Assume that (i) `rX` represents the register pointing to the beginning of the locations where the variables are stored, and (ii) the offsets of the variables in the program above are given in the table below. In the table, each capital letter represents an integer.

variable	offset
x	X
y	Y
z	Z

Which one of the following LC-3 instruction sequences will be generated when converting statement "y = z[2]" to LC-3 assembly language instructions?

- a. ADD R0, R5, #Z
  LDR R1, R0, #2
  STR R1, R5, #Y
- b. LDR R0, R5, #Z
  LDR R1, R0, #2
  STR R1, R5, #Y
- c. ADD R0, R5, #Z
  LDR R1, R0, #2
  LDR R2, R1, #0
  STR R2, R5, #Y
- d. LDR R0, R5, #Z
  ADD R1, R0, #2
  STR R1, R5, #Y

Your answer is correct.

The correct answer is:
ADD R0, R5, #Z
LDR R1, R0, #2
STR R1, R5, #Y

Question **10**

Correct

Mark 0.15 out of 0.15

Read the program and the descriptions below before answering the next question.

```
#include <stdio.h>
int main() {
  char* x;
  char y;
  char z[4] = "abc";
  x = z;
  y = z[2];
  z[0] = *(x+1);
  printf("%s\n",z);
}
```

Assume that (i) `rX` represents the register pointing to the beginning of the locations where the variables are stored, and (ii) the offsets of the variables in the program above are given in the table below. In the table, each capital letter represents an integer.

variable	offset
x	X
y	Y
z	Z

Which one of the following LC-3 instruction sequences will be generated when converting statement "z[0] = *(x+1)" to LC-3 assembly language instructions?

a.
```
LDR  R0, R5, #X
ADD  R1, R0, #1
LDR  R2, R5, #Z
STR  R1, R2, #0
```

b.
```
LDR  R0, R5, #X
ADD  R1, R0, #1
ADD  R2, R5, #Z
STR  R1, R2, #0
```

c.
```
LDR  R0, R5, #X
LDR  R1, R0, #1
ADD  R2, R5, #Z
STR  R1, R2, #0
```

d.
```
LDR  R0, R5, #X
ADD  R1, R0, #1
STR  R1, R5, #Z
```

Your answer is correct.

The correct answer is:

```
LDR    R0, R5, #X
LDR    R1, R0, #1
ADD    R2, R5, #Z
STR     R1, R2, #0
```

◄ Practice MCQ8 (for practice only)

Jump to...

Practice MCQ 9 (for practice only) ►